A one-cell deletion proof for bounded-excess growth of minimizing polyhexes

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Abstract

This note records a second proof of the one-cell growth theorem of the author's paper *Shortest-side skeletons of edge-perimeter-minimizing polyhexes*. Instead of deleting a whole extreme line at a time, it deletes one cell at a time, keeping the configuration directionally convex with at most one occupied line above the threshold and tracking two competing budgets through an admissibility condition. The existence of a legal deletion and the preservation of the condition are proved separately; the former rests on a bridge-excess lemma, the latter on a bound at the start of each plateau of the threshold, whose two exceptional cases are the rigidity statement used by the main paper. Two configurations show that neither auxiliary condition can be dropped. The common framework of the main paper is reproduced for self-containment.

1 The hexagonal lattice, runs, and directional convexity

This part reproduces, for self-containment, the common framework of the main paper [Utsunomiya, *Shortest-side skeletons of edge-perimeter-minimizing polyhexes*]: the lattice, the perimeter functional, directional convexity, the tight hull and its identities, and the extremal capacity function. Numbering follows this note.

1.1 Lattice and polyhexes

Cells of the hexagonal tiling are indexed by axial coordinates $(q, r) \in \mathbb{Z}^2$. We call q, r and

$$s := q + r$$

the three *keys* of a cell; in words, the third key of a cell is the sum of its two axial coordinates. Two cells are *adjacent* when they differ by one of the six vectors

$$d_0 = (1, 0), \quad d_1 = (0, 1), \quad d_2 = (-1, 1), \quad d_3 = (-1, 0), \quad d_4 = (0, -1), \quad d_5 = (1, -1).$$

A *polyhex* is a finite nonempty set of cells that is connected with respect to adjacency. We write $|S|$ for the number of cells of S.

The following four facts are immediate from the definitions and are used throughout without further comment.

Fact 1.1. Two distinct cells are adjacent if and only if they agree in exactly one key and differ by exactly 1 in the other two. In words: an adjacency step keeps one of the three keys fixed and changes each of the other two by one unit (the step $d_0 = (1, 0)$, for instance, sends (q, r, s) to $(q + 1, r, s + 1)$, so the two changing keys need not move in opposite senses).

Proof. Each d_j changes (q, r, s) by $(1, 0, 1)$, $(0, 1, 1)$, $(-1, 1, 0)$, $(-1, 0, -1)$, $(0, -1, -1)$ or $(1, -1, 0)$, and these are exactly the six triples with one zero entry and two entries equal to ± 1 that are realized. Conversely a difference of that shape is one of the d_j. □

Fact 1.2. A *line* is a set of cells on which one key is constant. Two lines belonging to different keys meet in exactly one cell; two distinct lines belonging to the same key are disjoint.

Fact 1.3. One adjacency step changes each of q, r and s by at most 1.

Fact 1.4. For a polyhex S and each key, the set of values taken by that key on S is an interval of consecutive integers. In words: no key value inside the range of S can be skipped.

Proof. Let k be a key and let $x, y \in S$ realize two values of k. A path of adjacent cells inside S joins them, and by Fact 1.3 the value of k changes by at most one per step, so every intermediate integer occurs on the path. □

1.2 Edge perimeter, runs and directional convexity

Let $a(S)$ denote the number of unordered pairs of adjacent cells of S. The *edge perimeter* of S is

$$p(S) = 6|S| - 2a(S), \tag{1}$$

in words: six times the number of cells, minus twice the number of adjacent pairs. Equivalently, $p(S)$ counts the cell edges of S that are not shared with another cell of S; we call these the *boundary edges*. The *edge semiperimeter* is

$$\sigma(K) := \tfrac{1}{2}p(K)$$

for any finite set of cells K; in words, half the edge perimeter. That this is an integer is part of Lemma 1.6 below.

A line Λ is *occupied* by S when $S \cap \Lambda \neq \emptyset$; we call $S \cap \Lambda$ the *trace* of S on Λ. Cells of a line are linearly ordered (for a line of constant q by the value of r, for one of constant r by the value of q, for one of constant s by the value of q), and consecutive cells in that order are adjacent. A *run* of S on Λ is a maximal block of consecutive cells of the trace. Let

$$R(S) = \text{the total number of runs of } S \text{ over all lines}, \qquad \lambda(S) = \text{the number of occupied lines}.$$

Definition 1.5. A polyhex S is *directionally convex* when the trace of S on every occupied line is a single run. In words: along each of the three directions, S never leaves a line and comes back to it.

This is a lattice notion and not Euclidean convexity; it is the notion called *layer convexity* in parts of the literature on lattice animals.

Lemma 1.6. *For every finite set of cells S,*

$$p(S) = 2R(S) \;\geq\; 2\lambda(S), \tag{2}$$

in words: the edge perimeter equals twice the total number of runs, and the total number of runs is at least the number of occupied lines. In particular $p(S)$ is even and $\sigma(S) = R(S)$. For a polyhex, equality $R(S) = \lambda(S)$ holds if and only if S is directionally convex.

Proof. Let $e = \{x, y\}$ be a boundary edge, with $x \in S$ and $y \notin S$ the two cells it separates. By Fact 1.1 the cells x and y agree in exactly one key, hence they are consecutive cells of exactly one line Λ, and x is an end of the run of S on Λ containing it. Conversely a run has exactly two ends, and the cell immediately beyond each end lies outside S and determines a boundary edge; for a run consisting of a single cell the two ends are the two opposite sides of that cell, so they give two distinct boundary edges. This is a bijection between boundary edges and ends of runs, whence $p(S) = 2R(S)$.

Every occupied line carries at least one run, so $R(S) \geq \lambda(S)$, with equality exactly when every occupied line carries exactly one run, which is Definition 1.5. □

Remark 1.7 (edge perimeter versus site perimeter). The *site perimeter* of S is the number of cells outside S adjacent to at least one cell of S. It is a different functional from (1), and the two have different minimizers in general; results about one do not transfer to the other. Nothing in this paper depends on the site perimeter. We mention it only because part of the literature we cite in Section 2.2 is written in those terms.

1.3 Extreme support lines and traces

Let S be a polyhex. For each key, Fact 1.4 makes the occupied values an interval, so there is a smallest and a largest occupied line in each of the three families. These six lines are the *extreme lines* of S; we list them in the cyclic order

$$\Lambda_1 :\ q = q_{\min}, \quad \Lambda_2 :\ r = r_{\max}, \quad \Lambda_3 :\ s = s_{\max}, \quad \Lambda_4 :\ q = q_{\max}, \quad \Lambda_5 :\ r = r_{\min}, \quad \Lambda_6 :\ s = s_{\min},$$

so that Λ_i and Λ_{i+3} are the two extreme lines of one and the same family, indices being read in $\mathbb{Z}/6$. If a family occupies a single line then the two corresponding extreme lines coincide; all statements below are read with that degeneracy allowed.

Definition 1.8. Let S be a *directionally convex* polyhex. Its six *support traces* are the traces $S \cap \Lambda_i$, each a single run by Definition 1.5, and we write

$$r_i := |S \cap \Lambda_i| \ \geq 1, \qquad i \in \mathbb{Z}/6, \qquad \text{and} \qquad \rho(S) := \min_i r_i.$$

In words: r_i is the number of cells of S on the i-th extreme line, and $\rho(S)$ is the smallest of those six numbers.

Remark 1.9. Definition 1.8 is stated for directionally convex S only. On that domain $\rho(S)$ is at once the length of the shortest support trace and the length of the shortest extreme-line run. For a general polyhex a trace need not be a run, and we do not extend the definition.

2 Tight hulls and exact identities

2.1 Tight lattice hulls

Definition 2.1. Let $L \geq 1$ and $u, v, w \geq 0$ be integers satisfying the *admissibility* conditions

$$u + v \leq L - 1, \qquad v + w \leq L - 1, \qquad u + w \leq L - 1, \tag{3}$$

and put $A := L - u$, $B := L - v$, $C := L - w$, so that $A, B, C \geq 1$. The corresponding *lattice hexagon* is

$$H(L; u, v, w) := \{\, (q, r) :\ 0 \leq q \leq A - 1,\ \ 0 \leq r \leq B - 1,\ \ w \leq q + r \leq L - 1 \,\}.$$

In words: the triangle of side L with three corner triangles of sides u, v and w cut off. Degenerate shapes, in which one or more of the cut corners is empty or one of the three widths equals 1, are allowed.

Definition 2.2. The *tight hull* $H(S)$ of a polyhex S is the lattice hexagon determined by the extreme values of the three keys on S. Concretely, normalize by a translation so that $q_{\min} = r_{\min} = 0$ and set

$$A = 1 + q_{\max}, \quad B = 1 + r_{\max}, \quad L = 1 + s_{\max}, \quad w = s_{\min}, \quad u = L - A, \quad v = L - B.$$

Lemma 2.3 (the tight hull is admissible)**.** *For every polyhex S, the parameters L, u, v, w of Definition 2.2 satisfy $L \geq 1$, $u, v, w \geq 0$ and the three inequalities* (3)*, so $H(S) = H(L; u, v, w)$ is a lattice hexagon.*

Proof. After the normalization every cell has $q, r \geq 0$, hence $s \geq 0$, so $w \geq 0$ and $L \geq 1$; and $u = s_{\max} - q_{\max}$, $v = s_{\max} - r_{\max}$. A cell with $q = q_{\max}$ has $s \geq q_{\max}$, so $u \geq 0$; likewise $v \geq 0$. A cell with $s = s_{\max}$ has $s_{\max} = q + r \leq q_{\max} + r_{\max}$, which is $u + v \leq L - 1$. A cell with $q = 0$ has $s = r \leq r_{\max}$, so $s_{\min} \leq r_{\max}$, which is $v + w \leq L - 1$; a cell with $r = 0$ gives $s_{\min} \leq q_{\max}$, which is $u + w \leq L - 1$. □

For a lattice hexagon H we write $\Lambda_1, \ldots, \Lambda_6$ for its six extreme lines in the cyclic order of Section 1.3, namely

$$q = 0, \quad r = B - 1, \quad s = L - 1, \quad q = A - 1, \quad r = 0, \quad s = w,$$

and call them the *side lines*; we write

$$\ell_i := |H \cap \Lambda_i|$$

for the six *side lengths*, $h := \min_i \ell_i$ for the shortest side length, and $\kappa_i := H \cap \Lambda_i \cap \Lambda_{i+1}$ for the *corner* cells. Finally we set

$$\delta(H) := \ell_1 - \ell_4, \qquad Q(H) := (\ell_1 - \ell_3)^2 + (\ell_3 - \ell_5)^2 + (\ell_5 - \ell_1)^2.$$

Lemma 2.4. *Let S be a polyhex and $H = H(S)$. Then*

(i) $S \subseteq H$;

(ii) S and H occupy exactly the same lines, so $\lambda(S) = \lambda(H)$;

(iii) each of the six side lines of H meets S.

Proof. (i) is immediate from Definition 2.2, since every cell of S satisfies the six inequalities defining H. For (ii), the lines occupied by S in the q-family are, by Fact 1.4, exactly $q = 0, \ldots, A - 1$, and likewise $r = 0, \ldots, B - 1$ and $s = w, \ldots, L - 1$; these are precisely the lines occupied by H, as read off from Definition 2.1. Assertion (iii) is the case of (ii) for the six extreme values: each of them is attained by some cell of S. □

Lemma 2.4 is the only place where connectedness of S is used in this section, and it is used through Fact 1.4: for a disconnected set the occupied key values need not form an interval, and (ii) fails.

2.2 Hull identities

Lemma 2.5. *Let $H = H(L; u, v, w)$ be a lattice hexagon. Then:*

(i) the six side lengths are

$$\ell_1 = L - v - w, \quad \ell_2 = v + 1, \quad \ell_3 = L - u - v, \quad \ell_4 = u + 1, \quad \ell_5 = L - u - w, \quad \ell_6 = w + 1,$$

and each side line meets H in a single run whose two end cells are the corners κ_{i-1} and κ_i;

(ii) H is a directionally convex polyhex with

$$\lambda(H) = \sigma(H) = A + B + C = 3L - (u + v + w);$$

(iii) the six side lengths satisfy

$$\sum_{i=1}^{6} \ell_i = \lambda(H) + 3, \tag{4}$$

in words: the six side lengths of the hull add up to three more than the number of lines the hull occupies; and the three differences across opposite sides agree,

$$\ell_1 - \ell_4 = \ell_3 - \ell_6 = \ell_5 - \ell_2 = \delta(H) = L - u - v - w - 1; \tag{5}$$

(iv) with $T(k) := k(k+1)/2$,

$$|H| = T(L) - T(u) - T(v) - T(w);$$

(v) (hull defect identity)

$$\lambda(H)^2 + 3 - 12|H| \;=\; 3\,\delta(H)^2 + 2\,Q(H) \;\geq\; 0, \tag{6}$$

in words: the square of the number of occupied lines, plus three, minus twelve times the number of cells, equals three times the square of the common opposite-side difference plus twice the imbalance $Q(H)$.

Proof. (i) On the line $q = 0$ the defining inequalities reduce to $w \leq r \leq \min(B-1, L-1) = B-1$, which is a nonempty interval because $w \leq B-1$ is the condition $v + w \leq L-1$ of (3); its length is $B - w = L - v - w$. On the line $r = B-1$ they reduce to $\max(0, w-B+1) \leq q \leq \min(A-1, v)$, and (3) gives $w - B + 1 \leq 0$ and $v \leq A-1$, so the interval is $0 \leq q \leq v$, of length $v+1$. The remaining four sides are obtained in the same way, each time using exactly one of the three inequalities of (3) at each end. Each side is an interval of consecutive cells of its line, hence a run; its two end cells are the two cells where a second defining inequality becomes an equality, that is, the two corners.

(ii) The occupied lines are $q = 0, \ldots, A-1$, $r = 0, \ldots, B-1$ and $s = w, \ldots, L-1$, which number $A + B + C$; on each of them the trace is an interval of consecutive cells by the same computation as in (i), so H is directionally convex and Lemma 1.6 gives $\sigma(H) = R(H) = \lambda(H)$. Connectedness follows since each trace is a run and consecutive traces meet a common line.

(iii) Summing the six expressions of (i),

$$\sum_i \ell_i = 3L - (u + v + w) + 3 = \lambda(H) + 3.$$

Using the side-length formulas of (i), the three opposite-side differences $\ell_1 - \ell_4$, $\ell_3 - \ell_6$, $\ell_5 - \ell_2$ of (5) are $(L-v-w)-(u+1)$, $(L-u-v)-(w+1)$ and $(L-u-w)-(v+1)$, all equal to $L-u-v-w-1$.

(iv) The triangle $\{q, r \geq 0,\ q+r \leq L-1\}$ has $T(L)$ cells. The three sets removed in Definition 2.1 are the corner triangles $q \geq A$, $r \geq B$ and $q+r \leq w-1$, of $T(u)$, $T(v)$ and $T(w)$ cells respectively, and (3) makes them pairwise disjoint: a cell with $q \geq A$ and $r \geq B$ would have $q+r \geq A+B = 2L-u-v \geq L+1$, which is impossible, and a cell with $q \geq A$ and $q+r \leq w-1$ would give $r \leq w-1-A < 0$, and symmetrically.

(v) Put $U := u+v+w$ and $P := u^2+v^2+w^2$. By (ii) and (iv),

$$\lambda(H)^2 + 3 - 12|H| = (3L-U)^2 + 3 - 6L(L+1) + 6P + 6U = 3L^2 - 6LU - 6L + U^2 + 6P + 6U + 3.$$

On the other side, $\delta(H) = L-U-1$ and, by (i), $Q(H) = (u-w)^2 + (w-v)^2 + (v-u)^2 = 3P - U^2$, so

$$3\delta(H)^2 + 2Q(H) = 3(L-U-1)^2 + 2(3P-U^2)$$

expands to the same expression. In words: both sides equal three times L^2, minus six times LU, minus six times L, plus U^2, plus six times P, plus six times U, plus three. Nonnegativity is clear from the right-hand side. □

Remark 2.6 (attribution). An equivalent defect identity is obtained from Fülep and Sieben's polyiamond identity [2, Lemma 3.3] through the triangular–hexagonal duality developed in their Section 5. In the same site-perimeter variables, [2, Proposition 5.10] yields exactly the capacity function $M(t)$ of Section 3.1 after the substitution $p = t+3$; the value $p = 7$ excluded there corresponds to the value $t = 4$ excluded in Remark 3.5 below. We nevertheless give the derivation here in edge-perimeter variables to keep the argument self-contained, and we do not identify the two settings: [2] is about the site perimeter (Remark 1.7).

3 Capacity and minimum edge perimeter

3.1 Extremal capacity

We first record the two operations on directionally convex polyhexes that the capacity computation needs: removing a whole support trace, and putting it back one cell at a time.

Lemma 3.1 (peeling). *Let K be a directionally convex polyhex, let Λ be one of its six extreme lines, and let $K' := K \setminus \Lambda$. If $K' \neq \emptyset$, then*

(i) K' is a directionally convex polyhex with $\sigma(K') \leq \sigma(K) - 1$;

(ii) the cells of $K \cap \Lambda$ can be ordered $c_1, \ldots, c_k$ so that for every j the set $K' \cup \{c_1, \ldots, c_j\}$ is a directionally convex polyhex of edge semiperimeter at most $\sigma(K)$.

Proof. Say Λ is the line $q = q_{\min}$; the other five cases are the same after permuting the roles of the keys and of the two senses.

(i) For each occupied value k of q let K_k be the trace of K on $q = k$; it is a run. If K is connected, then for consecutive values $k, k+1$ a path inside K joining a cell of K_k to a cell of K_{k+1} must, by Fact 1.3, contain a step from a cell with $q = k$ to a cell with $q = k+1$, so K_k and K_{k+1} are adjacent. Hence K', the union of the K_k with $k \geq q_{\min}+1$, is connected. Its traces in the q-family are traces of K, hence runs. Now fix a line of the r-family, $r = k$; its cells are (q, k), so the trace of K on it is a run if and only if its q-values form an interval, and by Fact 1.2 it meets Λ in at most the

one cell $(q_{\min}, k)$. The trace of K' on this line is the trace of K with that cell, if present, removed. Since $q_{\min}$ is the least q-value occurring anywhere in K, it is the least q-value of the interval, so removing it leaves an interval, possibly empty. A line of the s-family, $s = k$, has cells $(q, k - q)$, again parametrized by q, and the same sentence applies verbatim. So all traces of K' are runs and K' is directionally convex. Finally Λ is occupied by K and not by K', so $\lambda(K') \le \lambda(K) - 1$, and by Lemma 1.6 $\sigma(K') = \lambda(K') \le \lambda(K) - 1 = \sigma(K) - 1$.

(ii) Since K is connected there is a cell $c_1 \in K \cap \Lambda$ adjacent to K'. Order the remaining cells of the run $K \cap \Lambda$ so that every initial segment $\{c_1, \dots, c_j\}$ is a block of consecutive cells of that run containing c_1. Write $K_j := K' \cup \{c_1, \dots, c_j\}$. Then K_j is connected, because c_1 is adjacent to K' and each later c_j is adjacent to a cell already inserted. Its trace on Λ is a block, hence a run; its trace on any other line of the q-family is a trace of K', hence a run. On a line of the r- or s-family, parametrized by q as in (i), the trace of K_j is the trace of K' there together with at most one inserted cell, which lies on Λ and so has $q = q_{\min}$. The trace of K on that line is a run containing this cell and, by (i), the trace of K' is that run with its least element $q_{\min}$ removed; putting the cell back restores the run of K. Hence every trace of K_j is a run, so K_j is directionally convex, and every line it occupies is occupied by K, whence $\sigma(K_j) = \lambda(K_j) \le \lambda(K) = \sigma(K)$. □

Corollary 3.2. *Let K be a directionally convex polyhex. For every j with $1 \le j \le |K|$ there is a directionally convex polyhex $K_j \subseteq K$ with $|K_j| = j$ and $\sigma(K_j) \le \sigma(K)$.*

Proof. Strong induction on $|K|$. For $|K| = 1$ there is nothing to prove. Let $|K| \ge 2$. Some extreme line Λ does not contain all of K: otherwise K would be contained in two extreme lines belonging to different families, which by Fact 1.2 meet in a single cell, contradicting $|K| \ge 2$. Apply Lemma 3.1 to that Λ. The sets K_j of part (ii) supply all sizes between $|K'|$ and $|K|$; the sizes below $|K'|$ are supplied by the inductive hypothesis applied to K', whose edge semiperimeter is at most $\sigma(K)$. □

Definition 3.3. For an integer $t \ge 0$ put

$$M(t) := \left\lfloor \frac{t^2 + 3}{12} \right\rfloor,$$

in words: the integer part of one twelfth of the square of t plus three. Thus $M(0) = M(1) = M(2) = 0$ and $M(3) = 1$. The function M is nondecreasing.

Theorem 3.4. *Let $N \ge 1$.*

(a) Every polyhex S satisfies $|S| \le M(\sigma(S))$; in words, a configuration of edge semiperimeter t cannot have more than $M(t)$ cells.

(b) For every $t \ge 3$ there is a directionally convex polyhex K with $|K| = M(t)$ and $\sigma(K) \le t$.

(c) The minimum edge perimeter of an N-cell polyhex is

$$p_{\min}(N) = 2g(N), \qquad g(N) = \left\lceil \sqrt{12N - 3} \right\rceil,$$

in words: the smallest edge semiperimeter that can hold N cells is the least integer at least as large as the square root of twelve times N minus three, and the minimum edge perimeter is twice that number.

(d) Every N-cell minimizer S is directionally convex and satisfies $\sigma(S) = \lambda(S) = g(N)$.

(e) Writing $\operatorname{exc}(S) := p(S) - p_{\min}(|S|)$*, a polyhex* S *satisfies* $\operatorname{exc}(S) \le 2$ *if and only if* $|S| > M(\sigma(S) - 2)$.

The directional convexity in (d) is a known result: Davoli, Piovano and Stefanelli [1, Proposition 3.3] prove that every minimizer of the edge-isoperimetric problem on the triangular lattice is 3-convex, which is directional convexity in the present terminology; the proof below is the self-contained one of the main paper, via the hull identities.

Proof. (a) Let $H = H(S)$. By Lemma 2.4 and Lemma 2.5(v),

$$12|S| \le 12|H| = \lambda(H)^2 + 3 - \big(3\delta(H)^2 + 2Q(H)\big) \le \lambda(S)^2 + 3 \le \sigma(S)^2 + 3,$$

the last step by Lemma 1.6. Since $|S|$ is an integer, $|S| \le M(\sigma(S))$.

(b) For $t \equiv 3 \pmod 6$, say $t = 6m + 3$, take $K = H(3m + 1; m, m, m)$; for $t = 6m$ take $H(3m - 1; m - 1, m - 1, m - 1)$; for $t = 6m + 1$ take $H(3m; m, m, m - 1)$; for $t = 6m + 2$ take $H(3m + 1; m, m, m + 1)$; for $t = 6m + 4$ take $H(3m + 1; m, m, m - 1)$; for $t = 6m + 5$ take $H(3m + 2; m, m, m + 1)$. In each case (3) holds for the indicated range of m, Lemma 2.5(ii) gives $\sigma(K) = t$, and Lemma 2.5(iv),(v) give $12|K| = t^2 + 3 - D$ with $D \in \{0, 3, 4, 7\}$ according to the residue of t, so $|K| = M(t)$. The single case not covered by an admissible triple is $t = 4$; there $M(4) = 1$ and a single cell has edge semiperimeter $3 \le 4$.

(c) Put $t_N := \min\{t \ge 3 : \ M(t) \ge N\}$. By (a) every N-cell polyhex has $M(\sigma(S)) \ge N$, hence $\sigma(S) \ge t_N$. Conversely (b) produces a directionally convex K with $|K| = M(t_N) \ge N$ and $\sigma(K) \le t_N$, and Corollary 3.2 extracts from it an N-cell polyhex of edge semiperimeter at most t_N. Therefore $p_{\min}(N) = 2t_N$. Finally, for integers t,

$$M(t) \ge N \iff \frac{t^2 + 3}{12} \ge N \iff t \ge \sqrt{12N - 3},$$

and $\sqrt{12N - 3} \ge 3$ for $N \ge 1$, so $t_N = g(N)$.

(d) By (c), $\sigma(S) = g(N)$. Let $H = H(S)$. Applying (a) to H, which is directionally convex with $\sigma(H) = \lambda(H)$, gives $M(\lambda(H)) \ge |H| \ge N$ and therefore $\lambda(H) \ge t_N = g(N)$. Combining with Lemmas 1.6 and 2.4,

$$g(N) = \sigma(S) = R(S) \ \ge \ \lambda(S) = \lambda(H) \ \ge \ g(N),$$

so $R(S) = \lambda(S)$ and S is directionally convex.

(e) Write $t = \sigma(S)$ and $k = |S|$; by (c), $\operatorname{exc}(S) = 2\big(t - g(k)\big)$. Since both sides are integers, $\operatorname{exc}(S) > 2$ is equivalent to $g(k) \le t - 2$. If $t - 2 \ge 3$, then by the definition of g as t_k and the monotonicity of M this is equivalent to $M(t - 2) \ge k$. If $t - 2 < 3$, then $t \in \{3, 4\}$ because a polyhex has $t \ge 3$; here $M(t - 2) = 0 < k$ and $g(k) \ge 3 > t - 2$, so both statements are false. In both cases $\operatorname{exc}(S) > 2$ is equivalent to $M(t - 2) \ge k$, which is the assertion. □

Remark 3.5. Parts (a) and (b) together say exactly that

$$M(t) = \max\{\, |K| : \ K \text{ a polyhex with } \sigma(K) \le t \,\},$$

in words: $M(t)$ is the largest number of cells that fit within an edge-semiperimeter budget of t. The budget form is the one the rest of the paper uses, and the inequality in (b) is not a weakening: for every $t \ge 3$ other than $t = 4$ the hexagons listed in the proof do satisfy $\sigma(K) = t$, and the value $t = 4$ is not the edge semiperimeter of any polyhex at all. Indeed a polyhex K with $\sigma(K) = 4$ would have $|K| \le M(4) = 1$ by (a), while the one-cell polyhex has edge semiperimeter 3.

3.2 The plateau bridge

The function $N \mapsto g(N)$ is nondecreasing and takes each of its values on a block of consecutive integers. For γ in the range of g let

$$N_b(\gamma) := \min\{\, N \geq 1 :\ g(N) = \gamma \,\}$$

be the first cell count at which the minimum edge semiperimeter equals γ.

Lemma 3.6. *For every γ in the range of g,*

$$N_b(\gamma) = M(\gamma - 1) + 1, \tag{7}$$

in words: the plateau on which the minimum edge semiperimeter equals γ begins one cell after the capacity of edge semiperimeter $\gamma - 1$ has been exhausted.

Proof. For integers $N \geq 1$ and $t \geq 3$ we have, as in the proof of Theorem 3.4(c), $g(N) \leq t$ if and only if $N \leq M(t)$. Hence $\{N :\ g(N) \leq \gamma - 1\} = \{1, \ldots, M(\gamma - 1)\}$, so the least N with $g(N) \geq \gamma$ is $M(\gamma - 1) + 1$; as γ lies in the range of g, that least N satisfies $g(N) = \gamma$. □

4 Corridor and admissibility

The main paper proves the growth theorem by deleting a whole extreme line at a time and staying inside the minimizers. The proof recorded here descends one cell at a time and controls the result through a pair of competing budgets. It reads its geometry from the framework above, and it shares with the main paper the rigidity statement at exceptional plateau starts (Proposition 6.1 below, of which the main paper uses the two exceptional cases).

4.1 The lower bound on the line count

Lemma 4.1. *Every polyhex S satisfies $\lambda(S) \geq g(|S|)$.*

Proof. Let $H = H(S)$. By Lemma 2.5(ii) the hull is directionally convex with $\sigma(H) = \lambda(H)$, so Theorem 3.4(a) applied to H gives $|H| \leq M(\lambda(H))$. By Lemma 2.4, $S \subseteq H$ and $\lambda(H) = \lambda(S)$; hence $|S| \leq M(\lambda(S))$, and the definition of g as the least t with $M(t) \geq |S|$ gives $\lambda(S) \geq g(|S|)$. □

In words: the number of occupied lines can never fall below the threshold, because the hull would then be too small to hold the configuration.

4.2 Excess, band and corridor

Definition 4.2. For a polyhex S the *line-count excess* is

$$\varepsilon(S) := \lambda(S) - g(|S|)\ \geq 0,$$

in words: the number of occupied lines above the threshold. We set

$$\mathcal{B}_2 := \{\, S : \mathrm{exc}(S) \leq 2 \,\}, \qquad \mathcal{C} := \{\, S :\ S \text{ directionally convex},\ \varepsilon(S) \in \{0, 1\} \,\},$$

and call $\mathcal{B}_2$ the *2-excess band* and $\mathcal{C}$ the *unit line-excess corridor*, or simply the *corridor*.

If S is directionally convex then $p(S) = 2\lambda(S)$ by Lemma 1.6, and $p_{\min} = 2g$ by Theorem 3.4(c), so

$$\mathrm{exc}(S) = 2\,\varepsilon(S), \tag{8}$$

in words: for a directionally convex configuration the edge-perimeter excess is exactly twice the line-count excess. Hence $\mathcal{C} \subseteq \mathcal{B}_2$, and the inclusion is strict: the band also contains configurations that are not directionally convex, an explicit nine-cell one being given in Section 8.

4.3 Thresholds, plateaux and budgets

The threshold g is nondecreasing; write its increment as the *threshold jump*

$$j(N) := g(N) - g(N-1).$$

For $N \geq 3$ one has $j(N) \in \{0, 1\}$. (The value $j(2) = 5 - 3 = 2$ is exceptional; the iteration below is carried out for sizes at least 4.)

The function g is not surjective: $g(1) = 3$ and $g(2) = 5$, so $\gamma = 4$ is not a value of g. For γ in the range of g we use the plateau start $N_b(\gamma) = \min\{n : g(n) = \gamma\}$ of Section 3.2, for which Lemma 3.6 gives the closed form

$$N_b(\gamma) = M(\gamma - 1) + 1. \tag{9}$$

How far S has advanced along its plateau is the *plateau depth*

$$\tau(S) := |S| - N_b\big(g(|S|)\big) \ \geq 0.$$

(The symbol τ is used in this part for the plateau depth only; edge semiperimeters are written σ throughout, as in the framework above.)

For a directionally convex polyhex we write $r_1, \ldots, r_6$ for the support traces and $\rho(S) = \min_i r_i$ as in Definition 1.8; $\rho(S) \geq 1$ because no extreme line is empty. Every configuration to which ρ is applied below is verified to be directionally convex first, as Definition 1.8 requires.

4.4 The effect of a single deletion

Let $c \in S$ and $S' := S \setminus \{c\}$. Let $\mathrm{solo}(S, c)$ denote the *line-loss count*, the number of occupied lines that disappear when c is deleted. No new line is occupied, so $\lambda(S') = \lambda(S) - \mathrm{solo}(S, c)$, and Definition 4.2 gives at once

$$\varepsilon(S') = \varepsilon(S) + j(|S|) - \mathrm{solo}(S, c), \tag{10}$$

in words: deleting a cell raises the line-count excess by the threshold jump and lowers it by the number of lines vacated. This identity uses neither connectedness nor convexity. If S is connected with $|S| \geq 2$ then c has a neighbour, and each of the six directions of adjacency preserves one of the three keys, so $\mathrm{solo}(S, c) \leq 2$.

4.5 The admissibility condition

We wish to keep $\varepsilon \leq 1$ at every step. By (10) with solo ≥ 0 and $j \leq 1$, the value $\varepsilon(S')$ can reach 2 only when

$$\varepsilon(S) = 1, \qquad j(|S|) = 1, \qquad \mathrm{solo}(S, c) = 0$$

all hold at once: one spare line is already in use, the threshold drops as the size decreases, and no line is vacated to pay for it. The invariant of this part exists only to forbid that combination.

The device is a comparison of two budgets. The plateau depth $\tau(S)$ says how many further deletions are possible before the threshold drops again. The shortest support trace $\rho(S)$ says how many deletions are needed to exhaust the shortest extreme line, one cell at a time; at the deletion that exhausts it the line itself disappears, so solo ≥ 1 and the term $+j$ in (10) is cancelled. So $\rho(S) \leq \tau(S)$ is exactly the statement that *the line runs out before the threshold drops.*

Definition 4.3 (admissibility condition)**.** For a directionally convex polyhex S,

$$\mathrm{Adm}(S) \quad :\Longleftrightarrow \quad \varepsilon(S) = 0 \quad \text{or} \quad \big(\varepsilon(S) = 1 \text{ and } \rho(S) \leq \tau(S)\big).$$

Lemma 4.4. *If* $\mathrm{Adm}(S)$ *then* $\varepsilon(S) = 1$ *and* $j(|S|) \geq 1$ *cannot hold simultaneously.*

Proof. Assume $j(|S|) \geq 1$. As g is nondecreasing, $n < |S|$ gives $g(n) \leq g(|S| - 1) < g(|S|)$, so $|S|$ is itself the least size at which g takes the value $g(|S|)$, that is $\tau(S) = 0$. If $\varepsilon(S) = 1$, the second alternative of Adm would give $\rho(S) \leq 0$, contradicting $\rho(S) \geq 1$. □

Lemma 4.5 (initialization)**.** *Every edge-perimeter minimizer* S *satisfies* $S \in \mathcal{C}$ *and* $\mathrm{Adm}(S)$.

Proof. By Theorem 3.4(d), S is directionally convex with $\lambda(S) = g(|S|)$, that is $\varepsilon(S) = 0$. The first alternative of Definition 4.3 applies. □

What remains is to show that such a deletion exists at all (Section 5) and that it preserves Adm (Section 7). These two are established separately and are not combined into a single statement; the separation is the point of this proof, and Section 8 exhibits a configuration for which existence holds while admissibility fails.

5 Legal one-cell deletion

Proposition 5.1. *Let* $S \in \mathcal{C}$ *with* $|S| \geq 4$. *Then there is a cell* c *which is an endpoint of a shortest extreme line of* S *and whose deletion leaves* S *connected.*

Such a cell is called a *legal deletion*. Note that a single cell is removed; this is not the block peeling of the main paper, and the two operations are not interchangeable.

5.1 The witness cell

Among the six extreme lines of S choose one attaining $\rho(S)$; if there are several, any one will do. The symmetry group of the lattice permutes the three families cyclically and exchanges the minimum and maximum side within a family, so we may assume the chosen line is $q = q_{\max}$. Let

$$c^* := \operatorname{argmax}\{\, r : (q_{\max}, r) \in S \,\}$$

be the cell of that line with the largest r; we call it the *witness cell*. It is an endpoint of the trace on the chosen extreme line.

5.2 Local structure at the witness cell

Lemma 5.2. *None of* $c^* + d_0$, $c^* + d_1$, $c^* + d_5$ *belongs to* S. *Consequently* $\deg_S(c^*) \leq 3$, *and the neighbours of* c^* *inside* S *are confined to the three directions* d_2, d_3, d_4, *which are consecutive in the cyclic order.*

Proof. Both $d_0 = (1, 0)$ and $d_5 = (1, -1)$ increase q, and c^* lies on $q = q_{\max}$. The direction $d_1 = (0, 1)$ preserves q and increases r, and c^* has the largest r on that line. □

Corollary 5.3. *In each of the three families,* c^* *is an endpoint of the trace on the line containing it.*

Proof. In the q-family the line is $q = q_{\max}$ and c^* attains the maximum of r on it. In the r- and s-families the free coordinate along a line is q, and c^* has $q = q_{\max}$, the largest value of q on all of S. □

Corollary 5.4. *If c^* is an articulation cell of S, then $\deg_S(c^*) = 2$ and its two neighbours lie in the directions d_2 and d_4.*

Proof. If $\deg_S(c^*) \le 1$ then $S \setminus \{c^*\}$ is connected. If $\deg_S(c^*) = 3$ the neighbours are $c^* + d_2, c^* + d_3, c^* + d_4$, and since d_2, d_3 and d_3, d_4 are adjacent pairs of directions, these three cells are joined to one another inside $S \setminus \{c^*\}$, so any path through c^* can be rerouted. If $\deg_S(c^*) = 2$ with directions $\{d_2, d_3\}$ or $\{d_3, d_4\}$ the same applies. Only $\{d_2, d_4\}$ remains. □

5.3 A degree-two articulation cell forces excess at least four

Lemma 5.5 (arithmetic lemma)**.** *Let $a, b \ge 1$ be integers with $(a, b) \ne (1, 1)$. Then $g(a) + g(b) \ge g(a + b + 1) + 1$. For $(a, b) = (1, 1)$ the statement fails.*

Proof. Put $A := g(a)$, $B := g(b)$. By definition $A \ge \sqrt{12a - 3}$, that is $12a \le A^2 + 3$, and likewise for b; hence $12(a+b+1) - 3 \le A^2 + B^2 + 15$, and by monotonicity of the ceiling of the square root, $g(a+b+1) \le \lceil \sqrt{A^2 + B^2 + 15}\, \rceil$. Since $A+B-1$ is a positive integer, $\lceil \sqrt{A^2 + B^2 + 15}\, \rceil \le A+B-1$ is equivalent to $A^2 + B^2 + 15 \le (A + B - 1)^2$, that is to $(A-1)(B-1) \ge 8$. Now g is nondecreasing with $g(1) = 3$ and $g(2) = 5$, so $A = 3$ if $a = 1$ and $A \ge 5$ if $a \ge 2$, and similarly for B; unless $a = b = 1$ we get $(A - 1)(B - 1) \ge 2 \cdot 4 = 8$. Finally $A = B = 3$ happens exactly for $a = b = 1$, where $g(1) + g(1) = 6 = g(3)$. □

Lemma 5.6 (Bridge-Excess Lemma)**.** *Let X be a polyhex with $|X| = N \ge 4$ having a degree-two articulation cell. Then $\mathrm{exc}(X) \ge 4$.*

Proof. Let c be such a cell. Every component of $X \setminus \{c\}$ contains a neighbour of c, and c has two neighbours, so there are at most two components; being an articulation cell there are at least two. Call them X_1, X_2 and put $n_i = |X_i|$, so $n_1 + n_2 = N - 1$. Counting adjacent pairs, $a(X) = a(X_1) + a(X_2) + 2$, whence

$$p(X) = 6N - 2a(X) = p(X_1) + p(X_2) + 2.$$

In words: the two pieces keep their own perimeters and the bridge contributes two further boundary edges. With $p(X_i) \ge 2g(n_i)$ and Lemma 5.5,

$$p(X) \;\ge\; 2g(n_1) + 2g(n_2) + 2 \;\ge\; 2\big(g(N) + 1\big) + 2 \;=\; p_{\min}(N) + 4.$$

Lemma 5.5 needs $(n_1, n_2) \ne (1, 1)$, which would mean $N = 3$. □

5.4 Proof of Proposition 5.1

Proof. Let $S \in \mathcal{C}$ with $|S| \ge 4$. By (8), $\mathrm{exc}(S) \le 2$. Choose the witness cell c^*. If it were an articulation cell then by Corollary 5.4 it would be one of degree two, and Lemma 5.6 would give $\mathrm{exc}(S) \ge 4$. Hence $S \setminus \{c^*\}$ is connected, and c^* is a legal deletion. □

Remark 5.7 (what the word "shortest" is for). The argument just given used the shortest extreme line only to choose c^*; the connectedness conclusion did not depend on that choice. Indeed a theorem (recorded in a separate note) shows, by the same local argument, that for $|S| \ge 4$ and $\mathrm{exc}(S) \le 2$ *no* endpoint of *any* extreme line is an articulation cell, whether or not S is directionally convex. The restriction to a shortest extreme line is needed for a different reason: it is what the budget bookkeeping of Section 7 tracks through $\rho(S)$. The two hypotheses of that theorem cannot be dropped: in the bent three-cell polyhex $\{(0,1), (0,2), (1,0)\}$, of excess 2, the cell $(0,1)$ is an endpoint of the extreme line $q = 0$ and its deletion disconnects the set; and in the four-cell polyhex $\{(0,1), (0,2), (0,3), (1,0)\}$, of excess 4, the same is true of $(0,1)$.

6 The plateau-start bound

Proposition 6.1 (Plateau-Start Bound)**.** *Let γ be in the range of g with $\gamma \geq 7$, and let S be an edge-perimeter minimizer with $|S| = N_b(\gamma)$. Then*

$$\rho(S) \;\leq\; N_b(\gamma) - N_b(\gamma - 1).$$

Informally: at the start of a plateau the shortest extreme line of a minimizer cannot be longer than the distance down to the start of the previous plateau. This is what carries admissibility across the moment at which the threshold drops.

6.1 Two special values

By (9), for an integer $h \geq 2$,

$$N_b(6h-3) = 3h^2 - 4h + 2, \qquad N_b(6h-1) = 3h^2 - 2h + 1, \tag{11}$$

since $\big((6h-4)^2+3\big)/12$ has integer part $3h^2-4h+1$ and $\big((6h-2)^2+3\big)/12$ has integer part $3h^2-2h$.

6.2 Excess of the hull sides

All hull material is taken from the framework above: the parametrization $H(L;u,v,w)$ of Definition 2.1, the side lines $\Lambda_1,\dots,\Lambda_6$ in cyclic order, the side lengths $\ell_1,\dots,\ell_6$ and corners κ_i of Lemma 2.5(i), the sum rule (4), the opposite-side difference $\delta(H)$ of (5), and the shortest side length $h = \min_i \ell_i$. Write

$$x_i := \ell_i - h \;\geq 0$$

for the *side excess*, the amount by which a hull side exceeds the shortest hull side.

Lemma 6.2 (Opposite-Pair Excess Lemma)**.** *If $\sum_i x_i \leq 2$ then $x_i = x_{i+3}$ for every i; in particular $\sum_i x_i$ is even, so $\sum_i x_i = 1$ is impossible, and if $\sum_i x_i = 2$ then exactly one pair of opposite sides has length $h+1$ and the other four sides have length h.*

Proof. Since the x_i differ from the ℓ_i by the constant h, (5) gives $x_1 - x_4 = x_3 - x_6 = x_5 - x_2 = \delta(H)$. If $\delta(H) \geq 1$ then $x_1, x_3, x_5 \geq 1$ and $\sum x_i \geq 3$; if $\delta(H) \leq -1$ the same holds for x_2, x_4, x_6. Both contradict the hypothesis, so $\delta(H) = 0$ and $\sum x_i = 2(x_1 + x_3 + x_5)$. □

Lemma 6.3 (a complete boundary forces equality with the hull)**.** *Let S be a directionally convex polyhex and $H = H(S)$. If $H \cap \Lambda_i \subseteq S$ for all six i, then $S = H$.*

Proof. Fix a value q_0. By Definition 2.1 the cells of H on the line $q = q_0$ are those with r in the interval $\big[\max(0, w - q_0),\ \min(B-1, L-1-q_0)\big]$. Its lower end has $r = 0$ or $q + r = w$, so it lies on Λ_5 or Λ_6; its upper end has $r = B-1$ or $q + r = L-1$, so it lies on Λ_2 or Λ_3. By hypothesis both ends lie in S, and the trace of S on this line is a run, so it contains everything between them. As q_0 was arbitrary, $H \subseteq S$. □

6.3 Proof of Proposition 6.1

Proof. Let $N = N_b(\gamma)$ with $\gamma \geq 7$. By Theorem 3.4(d), S is directionally convex with $\lambda(S) = \gamma$, and by Lemma 2.4(ii) $\lambda(H) = \gamma$ for $H = H(S)$. Since $r_i \leq \ell_i$ for each i, we have $\rho(S) \leq h$. The minimum of six numbers is at most their mean, and it is an integer, so (4) gives

$$\rho(S) \;\leq\; h \;\leq\; \Big\lfloor \frac{\sum_i \ell_i}{6} \Big\rfloor = \Big\lfloor \frac{\lambda(H)+3}{6} \Big\rfloor = \Big\lfloor \frac{\gamma+3}{6} \Big\rfloor =: \beta(\gamma). \tag{12}$$

First, $N_b(\gamma-1)$ is defined: $N \geq 4$ and N is the least size with $g = \gamma$, so $g(N-1) < \gamma$; since $j(N) \in \{0,1\}$ for $N \geq 3$ we get $g(N-1) = \gamma - 1$, so $\gamma - 1$ is in the range of g. Write $\gamma - 1 = 6k + t$ with $t \in \{0, \ldots, 5\}$. By (9), $N_b(\gamma) - 1 = M(\gamma - 1) = 3k^2 + kt + \lfloor (t^2+3)/12 \rfloor$, and $\lfloor (t^2+3)/12 \rfloor$ equals $0, 0, 0, 1, 1, 2$ for $t = 0, \ldots, 5$. Taking differences, and noting that $\gamma - 2 = 6k + (t-1)$ when $t \geq 1$ while $\gamma - 2 = 6(k-1) + 5$ when $t = 0$, gives the following table.

t	$\gamma \bmod 6$	$N_b(\gamma) - N_b(\gamma-1) - k$	$\beta(\gamma) - k$	$\beta(\gamma) - \big(N_b(\gamma) - N_b(\gamma-1)\big)$
0	1	0	0	0
1	2	0	0	0
2	3	0	1	**1**
3	4	1	1	0
4	5	0	1	**1**
5	0	1	1	0

Hence $\beta(\gamma) - \big(N_b(\gamma) - N_b(\gamma-1)\big) = 1$ exactly when $\gamma \equiv 3$ or $5 \pmod 6$, and $\beta(\gamma) = N_b(\gamma) - N_b(\gamma-1)$ otherwise.

Case 1: $\gamma \not\equiv 3, 5 \pmod 6$**.** Then $\beta(\gamma) = N_b(\gamma) - N_b(\gamma-1)$ and (12) gives the claim at once.

Case 2: $\gamma \equiv 3 \pmod 6$**.** Write $\gamma = 6h - 3$; from $\gamma \geq 7$ we get $h \geq 2$. Here $\beta(\gamma) = h$ and we must show $\rho(S) \leq h - 1$. Suppose $\rho(S) = h$. Then (12) forces $h = \min_i \ell_i = \beta(\gamma)$, while $\sum_i \ell_i = \gamma + 3 = 6h$; as all six sides are at least h, all six equal h. Solving $\ell_2 = v + 1 = \ell_4 = u + 1 = \ell_6 = w + 1 = h$ and $\ell_1 = L - v - w = h$ gives

$$u = v = w = h - 1, \qquad L = 3h - 2,$$

so by Lemma 2.5(iv)

$$|H| = T(3h-2) - 3\,T(h-1) = 3h^2 - 3h + 1.$$

By (11), $N = 3h^2 - 4h + 2$, so $|H| - N = h - 1 \geq 1$. But $\rho(S) = h$ forces $h \leq r_i \leq \ell_i = h$ for every i, that is, every side of the hull lies entirely in S; Lemma 6.3 then gives $S = H$, contradicting $|S| = N < |H|$.

Case 3: $\gamma \equiv 5 \pmod 6$**.** Write $\gamma = 6h - 1$, again $h \geq 2$. Here $\beta(\gamma) = h$ and we must show $\rho(S) \leq h - 1$. Suppose $\rho(S) = h$; as before $\min_i \ell_i = h$, while now $\sum_i \ell_i = \gamma + 3 = 6h + 2$, so $\sum_i x_i = 2$. By Lemma 6.2 exactly one pair of opposite sides has length $h + 1$. The three families are permuted cyclically by the 120° rotation of the lattice, so we may assume that pair is (ℓ_1, ℓ_4), the two sides of the q-family. Solving $\ell_4 = u + 1 = h + 1$, $\ell_2 = v + 1 = h$, $\ell_6 = w + 1 = h$ and $\ell_1 = L - v - w = h + 1$ gives

$$u = h, \qquad v = w = h - 1, \qquad L = 3h - 1$$

(one checks $\ell_3 = \ell_5 = h$ and $\lambda(H) = 3L - (u + v + w) = 6h - 1 = \gamma$), and

$$|H| = T(3h-1) - T(h) - 2\,T(h-1) = 3h^2 - h.$$

By (11), $N = 3h^2 - 2h + 1$, so again $|H| - N = h - 1 \geq 1$.

From $\rho(S) = h$ it follows that the four sides of length h, namely $\ell_2, \ell_3, \ell_5, \ell_6$, lie entirely in S. The long side Λ_1 is adjacent in the cyclic order to Λ_6 and Λ_2, with which it shares the corners κ_6 and κ_1; both corners therefore lie in S, and since the trace of S on Λ_1 is a run containing both ends, Λ_1 lies entirely in S as well. The opposite side Λ_4 is handled the same way through its neighbours Λ_3 and Λ_5. So all six sides lie in S, and Lemma 6.3 gives $S = H$, contradicting $|S| = N < |H|$. □

6.4 The form used in Section 7

Corollary 6.4. *Let $S \in \mathcal{C}$ with $|S| \geq 4$, $\varepsilon(S) = 0$ and $j(|S|) = 1$. Put $\gamma := g(|S|)$. Then $|S| = N_b(\gamma)$ and $\rho(S) \leq N_b(\gamma) - N_b(\gamma - 1)$.*

Proof. As S is directionally convex with $\varepsilon(S) = 0$, Lemma 1.6 and Theorem 3.4(c) give $p(S) = 2\lambda(S) = 2g(|S|) = p_{\min}(|S|)$, so S is a minimizer. From $j(|S|) = 1$ and monotonicity of g we get $g(n) < \gamma$ for $n < |S|$, that is $|S| = N_b(\gamma)$; and $|S| \geq 4$ gives $\gamma \geq g(4) = 7$. Apply Proposition 6.1. □

7 Preservation and the alternative proof

Proposition 7.1. *Let $S \in \mathcal{C}$ with $|S| \geq 4$ and $\mathrm{Adm}(S)$, let c^* be the witness cell of Section 5.1, and put $S' := S \setminus \{c^*\}$. Then $S' \in \mathcal{C}$ and $\mathrm{Adm}(S')$.*

Connectedness of S' is Proposition 5.1. What remains is directional convexity, $\varepsilon(S') \in \{0, 1\}$, and $\mathrm{Adm}(S')$.

7.1 Directional convexity is preserved

Lemma 7.2. *Let S be directionally convex and let $c \in S$ be, in each of the three families, an endpoint of the trace on the line containing it. Then $S \setminus \{c\}$ is directionally convex.*

Proof. In each family exactly one line has its trace changed. An unchanged trace is still a run. The changed trace is a run of which c is an end, so removing c leaves a run, possibly empty. □

By Corollary 5.3 the witness cell is an endpoint in all three families, so S' is directionally convex. In particular $\rho(S')$ is defined, which is what permits its use below.

7.2 Tracking the two budgets

Write $\gamma := g(|S|)$.

Lemma 7.3 (plateau depth after a deletion)**.** (i) *If $j(|S|) = 0$ then $\tau(S') = \tau(S) - 1$.* (ii) *If $j(|S|) = 1$ then $|S| = N_b(\gamma)$, $\tau(S) = 0$ and $\tau(S') = N_b(\gamma) - 1 - N_b(\gamma - 1)$.*

Proof. (i) Here $g(|S| - 1) = \gamma$, so $\tau(S') = (|S| - 1) - N_b(\gamma) = \tau(S) - 1$, and $N_b(\gamma) \leq |S| - 1$ gives $\tau(S') \geq 0$. (ii) Monotonicity gives $g(n) < \gamma$ for $n < |S|$, so $|S| = N_b(\gamma)$ and $\tau(S) = 0$; also $g(|S| - 1) = \gamma - 1$, whence $\tau(S') = (|S| - 1) - N_b(\gamma - 1)$. □

Lemma 7.4 (shortest extreme line decrement)**.** *Let S be directionally convex, let $c \in S$ lie on a shortest extreme line, and suppose $\rho(S) \geq 2$ and that $S \setminus \{c\}$ is directionally convex. Then $\rho(S \setminus \{c\}) = \rho(S) - 1$.*

Proof. First, the six extreme lines of S and of S' coincide as pairs (family, key). Fix a family and let K be the set of keys occurring on S. If the key of c is neither the minimum nor the maximum of K, removing c changes neither. If it is extremal, the corresponding line is an extreme line and carries at least $\rho(S) \geq 2$ cells, so another cell has the same key and the extremes are again unchanged. Next, for each extreme line Λ we have $|S' \cap \Lambda| = |S \cap \Lambda| - [\, c \in \Lambda \,]$. All six values are at least $\rho(S)$; the value on the shortest extreme line containing c drops to $\rho(S) - 1$, and every other drops by at most one, so stays at least $\rho(S) - 1$. □

Lemma 7.5. *If c lies on a shortest extreme line and* $\mathrm{solo}(S, c) = 0$*, then* $\rho(S) \geq 2$*.*

Proof. If $\rho(S) = 1$ the shortest extreme line containing c consists of c alone, so deleting c vacates that line and $\mathrm{solo}(S, c) \geq 1$. □

7.3 The case distinction

By $\mathrm{Adm}(S)$ and Lemma 4.4, the combination $\varepsilon(S) = 1$ and $j(|S|) = 1$ does not occur. Moreover (10) together with $\varepsilon(S') \geq 0$, which is Lemma 4.1 applied to S', gives

$$\mathrm{solo}(S, c^*) \;\leq\; \varepsilon(S) + j(|S|). \tag{13}$$

With $\varepsilon(S) \in \{0, 1\}$ and $j(|S|) \in \{0, 1\}$ there are three cases.

Case A: $\varepsilon(S) = 0$**,** $j(|S|) = 0$**.** By (13), solo $= 0$, so (10) gives $\varepsilon(S') = 0$ and $\mathrm{Adm}(S')$ holds by its first alternative.

Case B: $\varepsilon(S) = 0$**,** $j(|S|) = 1$**.** Here solo ≤ 1.

B1: solo $= 1$. Then $\varepsilon(S') = 0 + 1 - 1 = 0$; first alternative.

B2: solo $= 0$. Then $\varepsilon(S') = 1$ and the second alternative must be established. All hypotheses of Corollary 6.4 hold, so $\rho(S) \leq N_b(\gamma) - N_b(\gamma - 1)$. By Lemma 7.5, $\rho(S) \geq 2$, and S' is directionally convex by Section 7.1, so Lemma 7.4 gives $\rho(S') = \rho(S) - 1$. By Lemma 7.3(ii), $\tau(S') = N_b(\gamma) - 1 - N_b(\gamma - 1)$. Therefore

$$\rho(S') = \rho(S) - 1 \;\leq\; \big(N_b(\gamma) - N_b(\gamma - 1)\big) - 1 \;=\; \tau(S').$$

Note that the bound of Corollary 6.4, decreased by one, coincides *identically* with the plateau depth after the deletion. In the language of Section 4.5: what remains of the shortest-line budget allowed at the start of the plateau, after one unit has been spent, is exactly the plateau budget afterwards. The form of Adm is chosen to make this branch close.

Case C: $\varepsilon(S) = 1$**,** $j(|S|) = 0$**.** Here solo ≤ 1.

C1: solo $= 1$. Then $\varepsilon(S') = 1 + 0 - 1 = 0$; first alternative.

C2: solo $= 0$. Then $\varepsilon(S') = 1$. The second alternative of $\mathrm{Adm}(S)$ gives $\rho(S) \leq \tau(S)$; by Lemmas 7.5 and 7.4, $\rho(S') = \rho(S) - 1$, and by Lemma 7.3(i), $\tau(S') = \tau(S) - 1$. Hence $\rho(S') \leq \tau(S')$. This case does not use Section 6.

In every case $\varepsilon(S') \in \{0, 1\}$ and $\mathrm{Adm}(S')$; with Section 7.1 and Proposition 5.1 this gives $S' \in \mathcal{C}$.

Remark 7.6. What Proposition 7.1 constrains is the size *before* the deletion, so it applies at $|S| = 4$ and its conclusion holds for $|S'| = 3$; it cannot be applied again with S' in the role of S. This is where the iteration stops, and sizes at most 3 are handled directly below.

7.4 The one-cell deletion proof of the growth theorem

Lemma 7.7. *Every polyhex X with $|X| \le 3$ satisfies* $\mathrm{exc}(X) \le 2$.

Proof. For $|X| = 1$, $p = 6 = 2g(1)$. For $|X| = 2$ there is one adjacent pair, so $p = 12 - 2 = 10 = 2g(2)$. For $|X| = 3$ the number of adjacent pairs is 2 or 3, so $p \in \{12, 14\}$, and $p_{\min}(3) = 2g(3) = 12$. □

Lemma 7.8. *Every polyhex X with $|X| \ge 2$ has a cell whose deletion leaves X connected.*

Proof. The adjacency graph of X is connected with at least two vertices; a leaf of any spanning tree is not a cut vertex. □

Theorem 7.9 (one-cell growth)**.** *Let S be an N-cell edge-perimeter minimizer. Then there are polyhexes $S_1 \subset \cdots \subset S_N = S$, $|S_k| = k$, each obtained from the previous one by adding an adjacent cell, with* $\mathrm{exc}(S_k) \le 2$ *for every k.*

Proof. For $N \le 3$ apply Lemma 7.8 repeatedly, removing one cell at a time while keeping connectedness; every configuration on the way is a polyhex of size at most 3, so (b) holds by Lemma 7.7, and (a) follows as in the reversal step below. Assume $N \ge 4$.

Initialization. By Lemma 4.5, $S \in \mathcal{C}$ and $\mathrm{Adm}(S)$. Put $T_N := S$.

Descent. Suppose $T_k \in \mathcal{C}$, $|T_k| = k \ge 4$ and $\mathrm{Adm}(T_k)$. By Proposition 5.1 there is an endpoint c_k of a shortest extreme line of T_k with $T_k \setminus \{c_k\}$ connected; put $T_{k-1} := T_k \setminus \{c_k\}$. By Proposition 7.1, $T_{k-1} \in \mathcal{C}$ and $\mathrm{Adm}(T_{k-1})$. Repeating down to $k = 4$ produces T_3 with $|T_3| = 3$; by Remark 7.6 the step at $k = 4$ is legitimate and the iteration stops there. Sizes at most 3 are continued with Lemma 7.8: choose $T_3 \supset T_2 \supset T_1$ preserving connectedness.

Staying in the band. For $k \ge 4$ we have $T_k \in \mathcal{C}$, so $\mathrm{exc}(T_k) \le 2$ by (8); for $k \le 3$ this is Lemma 7.7.

Reversal. Put $S_k := T_k$ and, for $2 \le k \le N$, let x_k be the unique cell of $T_k \setminus T_{k-1}$. Since T_k is connected with $|T_k| \ge 2$, the cell x_k has a neighbour inside T_k, necessarily in $T_{k-1} = S_{k-1}$. This is (a), and (b) was just shown. □

Corollary 7.10. *Let $\mathcal{R}_N$ be the class of N-cell polyhexes reachable from a single cell by a one-cell growth sequence all of whose intermediate configurations have edge-perimeter excess at most 2, and $\mathcal{M}_N$ the class of N-cell edge-perimeter minimizers. Then $\mathcal{M}_N \subseteq \mathcal{R}_N$ for every $N \ge 1$; that is, restricting candidate generation to one-cell growth with excess at most 2 excludes no minimizer. The reverse inclusion fails: $\mathcal{R}_N$ contains many non-minimizers.*

8 Two obstructions

The proof above runs its induction inside an auxiliary class rather than inside the class of minimizers, and pays for two devices that may look avoidable. Two explicit configurations show that they are not.

8.1 The corridor is a proper subset of the band

Example 8.1. Let

$$X_9 := \{(0,1), (0,2), (0,3), (1,0), (1,1), (1,2), (1,3), (2,0), (2,2)\}.$$

Then $|X_9| = 9$, $\lambda(X_9) = 11 = g(9)$ and $R(X_9) = 12$, so $p(X_9) = 24 = p_{\min}(9) + 2$ and $\mathrm{exc}(X_9) = 2$.

In words: the number of occupied lines is exactly the threshold, but one line carries two runs instead of one, so the configuration is not directionally convex while its edge-perimeter excess is still 2. All occupied lines have a single run except $q = 2$, whose trace $\{(2, 0), (2, 2)\}$ consists of two runs; thus $R(X_9) = \lambda(X_9) + 1$.

Hence $X_9 \in \mathcal{B}_2 \setminus \mathcal{C}$, and the inclusion $\mathcal{C} \subsetneq \mathcal{B}_2$ of Section 4.2 is strict. This is the reason the corridor is defined with directional convexity built in rather than by the excess bound alone: on $\mathcal{B}_2$ the identity $\mathrm{exc} = 2\varepsilon$ is unavailable, and the whole bookkeeping of the proof above rests on it.

8.2 The corridor alone is not enough: a fifteen-cell trap

Example 8.2. Let

$$X_{15} := \{(0, 4), (0, 5)\} \cup \{(1, r) : 1 \le r \le 5\} \cup \{(2, r) : 0 \le r \le 4\} \cup \{(3, r) : 0 \le r \le 2\}.$$

Then X_{15} is a directionally convex polyhex with $|X_{15}| = 15$, $\lambda(X_{15}) = 15$ and $g(15) = 14$, so $\varepsilon(X_{15}) = 1$ and $X_{15} \in \mathcal{C}$ with $\mathrm{exc}(X_{15}) = 2$. However

$$\rho(X_{15}) = 2, \qquad \tau(X_{15}) = 0,$$

so $\mathrm{Adm}(X_{15})$ fails, and every polyhex obtained from X_{15} by deleting one cell has excess at least 4.

In words: the shortest extreme line still carries two cells while the plateau budget is already exhausted, which is exactly the situation the admissibility condition forbids; and at that point no single deletion can keep the excess at 2.

Consequently the naive statement

> *every polyhex of excess at most* 2 *with at least two cells has a cell whose deletion leaves a polyhex of excess at most* 2

is false, and so is the version restricted to the corridor. The descent above therefore cannot be run on $\mathcal{C}$ alone: the pair $\mathcal{C} \cap \{\mathrm{Adm}\}$ is what closes, and Example 8.2 is the witness that the second condition is doing work. Note that this does not contradict Proposition 5.1, which asserts the existence of a connectedness-preserving deletion; here such deletions exist, but none of them stays in the band.

References

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